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Adopting AI-Enabled Solutions for Asset Optimization Towards Operational Excellence

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Abstract

Operational Excellence (OE) is now a strategic priority across the oil and gas industry, relying on effective Facility Management and Asset Performance Optimization, where sensors and models have long been pivotal. Yet in many existing facilities, the deployment of new sensors is constrained, and some asset components are too complex for conventional modeling tools. While Digital Transformation (DT) offers pathways to overcome these limitations, it is often viewed as costly and IT-driven, with limited engagement from asset engineers. Moreover, engineers trained in tools grounded in the physical sciences —such as hardware sensors and simulation platforms—often regard AI-enabled DT solutions, including soft sensors and ANN-based models, with skepticism. This paper addresses these challenges by presenting an innovative, versatile modeling framework that enables practical, resource-efficient digital solutions and encourages field personnel to apply AI-enabled tools to domain-specific problems in pursuit of Operational Excellence.

This paper presents the development of a versatile Artificial Neural Network (ANN)-based process model implemented using standard spreadsheet software (MS Excel), allowing practical deployment without the need for specialized platforms. A multilayer perceptron (MLP) architecture was employed to map five independent variables to one dependent output. The model's flexibility enables adaptation to diverse oilfield scenarios through updated input-output datasets. It was trained and tested on multiple datasets across various problem statements to demonstrate its adaptability, rapid responsiveness to input changes, and applicability to process modeling, soft sensing, and asset performance optimization in support of Operational Excellence.

The ANN-based model, developed using standard spreadsheet software, showed promising results when applied to various oilfield problems and datasets. Its predictions closely matched those from more advanced software tools. The model was able to capture complex relationships that are often difficult to represent using traditional methods, highlighting its potential for real-world applications. Because it is lightweight, user-friendly, and doesn't require specialized software or coding expertise, it can help more engineers feel comfortable using AI-based tools in their everyday work. This, in turn, can support a broader shift toward digital transformation in asset operations and promote more efficient progress toward Operational Excellence.

The paper promulgates an innovative concept of a versatile process model, implemented using ANN in spreadsheets, as a practical tool for facility management and asset optimization, thereby helping reduce barriers to the adoption of AI-enabled solutions for Operational Excellence.

Introduction

Operational Excellence (OE) is widely recognized today as a strategic imperative. However, its interpretation varies across organizations, influenced by differences in priorities, operational processes, and maturity levels in implementation. Treacy and Wiersema (1993), in Harvard Business Review, define Operational Excellence as "a specific strategic approach to the production and delivery of products and services. The objective of a company following this strategy is to lead its industry in price and convenience. Companies pursuing operational excellence are indefatigable in seeking ways to minimize overhead costs, to eliminate intermediate production steps, to reduce transaction and other 'friction' costs, and to optimize business processes across functional and organizational boundaries. They focus on delivering their products or services to customers at competitive prices and with minimal inconvenience." Nolan and Anderson (2015) characterize OE as "a journey that defines an organization and the capability it nurtures in pursuit of world class performance and competitive advantage." They note that "OE is largely about performance management and it is also a mindset. It helps keep everyone focused on doing the right things at the right time for the right reasons." In the context of oil, gas, and process industries, they define it more specifically as "the systematic management of health, environmental and safety in an integrated manner applied across all facets of the business to achieve superior outcomes across all operations of the organization." Similarly, Chevron Corporation (n.d.) defines OE as the systematic management of "workforce safety and health, process safety, reliability and integrity, environment, efficiency, security, and stakeholders." Accordingly, effective Facility Management (FM) and integrated Asset Performance Optimization (APO) are among the enabling elements of OE, as they contribute to maximizing production, reducing downtime, protecting people and the environment, and extending equipment life in a responsible and cost-effective manner.

Recent advancements in information and communication technologies (ICT) have created unprecedented opportunities for effective Facility Management and integrated Asset Performance Optimization, and thereby for attaining and sustaining OE. The integration of cyber-physical systems (CPS), the Industrial Internet of Things (IIoT), and Artificial Intelligence (AI) has significantly advanced traditional approaches to asset management by enabling predictive, adaptive, and integrated decision-making across the value chain. These technologies are widely recognized as key enablers of the Fourth Industrial Revolution (commonly referred to as Industry 4.0), which in turn contributes to OE. In the cyber-physical system, sensors and models play an important role. Sensors capture real-time operational data, while models interpret and predict asset performance. Together, they provide a foundation for monitoring, decision support, and optimization, which are essential components of OE. Notably, models themselves can function as sensors, commonly referred to as soft sensors, as further explained in this paper. However, in many facilities, timely deployment of new hardware sensors remains constrained due to a variety of reasons, such as unavailability of sufficient space in the existing system, operational unacceptability of shutting down the facility to carry out modifications, or funding constraints linked to approval cycles. Moreover, conventional modeling techniques often struggle to address the complexity and nonlinearity of real-world asset behavior. Digital Transformation (DT), particularly through AI-enabled solutions such as soft sensors and artificial neural networks (ANN), provides a promising pathway to address these limitations. Yet adoption is often hindered by skepticism from engineers trained in the physical sciences, by perceptions of high costs, and by the view that such initiatives are primarily IT-driven, with limited engagement from asset engineers.

This paper recommends adopting AI-enabled solutions for asset optimization towards attaining and sustaining Operational Excellence. It motivates asset engineers regarding AI-enabled solutions by presenting the development of a versatile model built upon the fundamental concept of Artificial Neural

Networks (ANN), from the rapidly advancing domain of Artificial Intelligence, using widely available in-house resources (standard spreadsheet software), and demonstrating its application for predicting Reid Vapor Pressure (RVP) at an oilfield facility, along with its significance to OE.

AI-enabled Digital Transformation for Operational Excellence

Digital Transformation (DT) is increasingly positioned as a critical enabler of Operational Excellence (OE), offering technologies that improve decision-making, enhance transparency, strengthen integration, and enable proactive actions in complex operations. Rogers (2016) emphasizes that DT is not merely about deploying technology but about rethinking organizational strategy, with influence across customers, competition, data, innovation, and value. In oil and gas, DT initiatives often include IoT networks for real-time asset visibility, digital twins that simulate facility operations, and cloud-based platforms that integrate data across the value chain. These technologies establish the digital foundation upon which AI-enabled models such as soft sensors, AI-driven optimization algorithms, and neural networks can be effectively applied to complement conventional sensors and state-of-the-art simulators. By doing so, these models provide predictive and adaptive capabilities that help organizations anticipate risks, respond to change, optimize performance, and support OE.

Opportunities

The integration of AI-enabled solutions into digitally transformed operations, as part of Industry 4.0 initiatives, presents significant opportunities for driving OE. Building on the foundation provided by IoT networks, digital twins, and cloud platforms, AI-enabled models expand organizational capacity to generate actionable insights and to achieve predictive, adaptive, and prescriptive management of assets. These models complement conventional sensors and simulators by uncovering hidden relationships in operational data and enabling real-time optimization of complex processes.

Predictive maintenance represents one of the most immediate opportunities. By applying machine learning models to sensor data and historical patterns, equipment faults can be anticipated and rectified, thereby reducing downtime and extending asset life (Hassan, Rossi, and Ivanova, 2019). Systematic reviews further emphasize that predictive models are being adopted across the energy sector, helping organizations transition from reactive to proactive maintenance strategies (Maroufkhani et al., 2022).

Another important opportunity lies in the application of models serving as soft sensors, which provide virtual estimates of parameters in situations where deployment of new hardware sensors is constrained. In oilfield operations, soft sensors are applied to predict variables such as vapor pressure, fluid composition, and equipment degradation, thereby complementing hardware instrumentation and improving monitoring reliability (Sircar et al., 2021).

AI-enabled models are also transforming production and process optimization. Artificial neural networks (ANN) and genetic algorithms capture nonlinear interactions among reservoir properties, flow rates, and pressures that are challenging to capture using conventional modeling approaches. These applications have been shown to improve drilling performance, reduce non-productive time, and enhance operational safety (Sircar et al., 2021; D'Almeida et al., 2022).

At a broader operational scale, AI-driven optimization models support improved scheduling, procurement, and supply chain management. Colli et al. (2020) highlight how IoT and automation, combined with AI analytics, enable greater resilience and efficiency in logistics. Digital twin models add another layer of opportunity by replicating facilities in virtual environments, allowing operators to test scenarios, evaluate risks, and optimize performance before implementing changes in the field (Neuhold, Kilger, and Poeschl, 2020).

Collectively, these opportunities illustrate the growing role of AI-enabled models as practical enablers of digital transformation in oil and gas. Whether applied as predictive maintenance tools, soft sensors, AI-driven optimization algorithms, or digital twins, such models convert large volumes of operational data into actionable insights that directly support OE objectives of reliability, safety, efficiency, and sustainability. For a more comprehensive understanding of these opportunities, Chelliah, Jayasankar, Agerstam, Sundaravadivazhagan, and Cyriac (2024) provide an extensive review of AI applications across subsurface analytics, production optimization, maintenance, environmental monitoring, and enterprise decision-making. In short, AI-enabled digital transformation offers significant potential for attaining and sustaining Operational Excellence, with AI models serving as key building blocks.

Challenges

Despite its potential, the adoption of AI-enabled digital transformation (DT) in the oil and gas sector has been limited. Several challenges persist, ranging from technical to organizational. One critical challenge is the constraint on hardware sensor deployment. Even when AI can offer soft-sensing alternatives, organizations often hesitate to rely on data-driven models over established physical measurements. Similarly, many asset components are too complex for conventional modelling approaches, but skepticism persists toward using artificial neural networks (ANN) or other AI tools to address this gap.

Another challenge lies in the reliance on state-of-the-art simulation platforms. While these tools are powerful, they demand expensive software licenses, high computational resources, and advanced expertise in specific disciplines. Their use often creates specialization silos, limiting cross-functional engagement across the value chain. This reinforces a perception that advanced modeling belongs to a narrow set of experts rather than a shared organizational capability, thereby limiting its broader utilization.

Cost perception further compounds these challenges. AI-enabled DT initiatives are often regarded as resource-intensive projects, requiring capital expenditure without assured short-term returns. This perception was reinforced by early digital projects that prioritized technology acquisition over operational outcomes, leaving frontline engineers skeptical about tangible value creation.

Cultural resistance presents another challenge. Engineers trained in the physical sciences often approach AI solutions with caution. Their expertise lies in phenomenological models and hardware sensors, and they may perceive ANN-based approaches as "black boxes" lacking clear physical correlations. Combined with organizational silos separating IT and engineering, this skepticism slows adoption and undermines the integration of AI-enabled solutions into daily operational practices.

Resolution

Overcoming these challenges requires a balanced approach that integrates technological innovation with cultural and organizational transformation.

First, adoption of AI-enabled solutions must be explicitly tied to OE objectives. ANN and soft-sensor models should be presented as tools for asset performance, safety, and sustainability rather than as stand-alone technology projects. This framing helps organizations shift the narrative from cost burden to strategic enabler. Demonstrating tangible benefits such as improved uptime, reduced emissions, and enhanced energy efficiency is essential for building trust and securing investment.

Second, democratization of AI tools provides a direct response to both specialization and cost challenges. Developing and deploying ANN-based models in MS Excel platforms offers a universal, low-cost, and discipline-agnostic solution once the model is established. Spreadsheets are already familiar across all levels of the organization, from engineers and operators to managers. Their use reduces dependence on expensive licenses, lowers computational overhead, and removes the specialization barrier imposed by conventional simulation platforms. This democratization makes integrated modeling across the value chain accessible to all, fostering inclusivity and collaboration that are essential for supporting OE.

Third, organizations must align human expertise with AI capabilities. Training and change-management programs can help engineers understand how AI complements rather than replaces physics-based and phenomenological models. Hybrid workflows, in which conventional models are supported by ANN-based predictive layers, can bridge interpretability with adaptability. Involving engineers in the development and deployment of ANN models can transform skepticism into ownership, embedding AI into daily practices of facility management and asset performance optimization.

Finally, an easily accessible versatile model developed in spreadsheets offers the opportunity to gain hands-on experience with an inexpensive and transparent AI-enabled solution. A practical example, such as using ANN to predict the Reid Vapor Pressure (RVP) of stabilized crude in spreadsheets, illustrates how AI can overcome physical measurement constraints and add value to OE initiatives. Such use-cases help build confidence, motivate staff participation, and reinforce the view that AI-enabled digital transformation provides a practical foundation for robust, enterprise-wide progress toward Operational Excellence.

Development of the Versatile Model (ANN-based)

The development of the Versatile Model (ANN-based) requires careful attention to several design factors that determine its accuracy, robustness, and ease of deployment. In the context of process industries, where operating conditions are nonlinear and multi-scale, these design choices become particularly critical for ensuring both reliability and practical applicability. Some of the core considerations in the development of such a model are discussed below:

ANN Type Selection

Artificial Neural Networks (ANN) can be classified in many ways, depending on topology, direction of data flow, and type of learning. A useful taxonomy is provided by Huang and Zhang (1994). The most general form is the Fully Connected Network, in which every node is connected to every other node in both directions. Although this architecture is theoretically comprehensive, it is seldom used in practice because the number of parameters grows very large with network size (n^2 weights for n nodes) (Mehrotra et al., 1997).

Another major class is the Layered Network, where nodes are organized into layers. If no intra-layer connections exist and information flows strictly forward, the network becomes an Acyclic Network. The simplest form of such networks, where connections occur only between successive layers (k to k+1), is termed a Feedforward Network. Conversely, when intra-layer or feedback connections exist, the network becomes Recurrent, leading to more complex computational behavior.

Beyond topology, ANNs can also be classified by their learning methods, most commonly supervised and unsupervised learning. Both methods comprise several sub-categories (Walczak and Cerpa, 1999). Over time, various types of artificial neural networks (ANNs) have been developed, including:

- Recurrent Neural Network (RNN)
- Hopfield Network
- Kohonen Network
- Radial Basis Function (RBF) network
- Multilayer Perceptron (MLP)

According to Meireles et al. (2003), practical applications at that time were dominated by the MLP (81.2%), with smaller shares for the Kohonen Network (8.3%), Hopfield Network (5.4%), and other types (5.1%). While newer architectures have since gained prominence, the MLP remains a strong candidate for the present work. The Multilayer Perceptron (MLP) is a feedforward network in which the outputs from one layer are connected to all nodes in the next layer. Literature across oil and gas applications confirms its versatility in handling nonlinear functions of varying complexity (Biyanto et al., 2007; García Esteban, 2009; Al Hutmany, 2013; Alshehri, 2009; Al-Otaibi et al., 2005). Vafaei et al. (2009) further affirm that the

MLP is the most widely adopted type for function approximation, with proven capability to approximate complex nonlinear processes. In view of its versatility, robustness, and ease of implementation, the present work adopts the MLP architecture.

Selecting the Topology of the MLP Network

Number of Inputs and Outputs. The topology of an MLP begins with identifying input and output variables. A priori knowledge of the process guides this selection. Also, emerging practices of data science can be used. However, too many input/output variables relative to the size of the training dataset may degrade learning quality and generalization ability. Rafiq et al. (2001) have listed some of the techniques to optimize the number of inputs and outputs.

Research also shows that it is not always necessary to develop a single multi-input, multi-output (MIMO) ANN for complex processes. Instead, Multiple Input, Single Output (MISO) models can be developed separately for each dependent variable. Such single-output ANNs often improve performance and simplify training. For example, Altissimi et al. (1998) used single-output ANNs in optimizing a separation plant, and Al-Otaibi et al. (2005) modeled dehydration and desalting process with separate ANNs for each output variable of interest.

Past studies report diverse input configurations: Al Hutmany (2013) used seven inputs, Chaudhuri and Ghosh (2009) used three, while Al-Otaibi et al. (2005) used five for desalting and dehydration process modeling. In line with such findings, and taking into account usability, five input variables were selected for the present model. This number was considered sufficient to capture relevant process behavior without overwhelming the model. Thus, a MISO configuration was selected for the versatile model.

Number of Hidden Layers. In most studies, a single hidden layer is considered sufficient (Hammer and Villmann, 2003). Many successful ANN applications in refining and separation processes have relied on a single hidden layer (Arce-Medina and Paz-Paredes, 2009; Al-Otaibi et al., 2005; Chaudhuri and Ghosh, 2009), though Al Hutmany (2013) experimented with two hidden layers. For the current development, one hidden layer was deemed sufficient and computationally efficient.

Number of Neurons in the Hidden Layer. The number of hidden neurons strongly influences model performance. Too few neurons lead to underfitting and oversimplification, while too many risk overfitting and excessive training time (Basheer and Hajmeer, 2000). Several heuristics exist, but trial-and-error remains the most widely accepted method for identifying optimal size. Reported industrial applications vary widely: Alshehri (2009) used 21 neurons; Chaudhuri and Ghosh (2009) used six; Al-Otaibi et al. (2005) found 10 neurons optimal through iteration. Arce-Medina and Paz-Paredes (2009) showed that even 3 neurons can perform well for the complex HDS process. Based on these precedents, the present work selected 10 neurons in a single hidden layer to achieve a balance between model accuracy and generalization. Thus, an MLP having five inputs, one output, one hidden layer with ten neurons, is selected for further model development. This configuration is shown schematically in Fig. 1.

In this figure, each circle represents an artificial neuron (node), arranged in three logical layers: input, hidden, and output. The input layer passes forward the five process variables without computation, while the hidden and output layers perform neural computations. Each connection between nodes carries an associated weight (e.g., w_{11}, w_{12} …), and every neuron also connects to a bias term (b_1, b_2, … b_{10}, and b in the output layer) (Hu and Hwang, 2002).

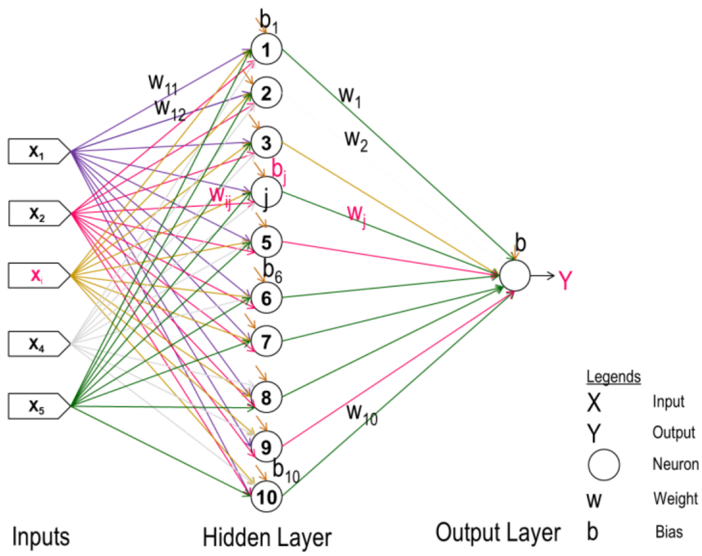

Figure 1—Proposed Artificial Neural Network for the Versatile Model

Input–Output Mapping in the MLP Model

The selected MLP provides a nonlinear mapping between its inputs (X_1–X_5) and the output Y. This mapping can be expressed mathematically, as summarized by Cameron and Hangos (2001), but is explained here step-by-step to highlight the computational power embedded in such networks.

Input Layer. For the sake of clarity interaction of single input (say X_3) with all nodes at the hidden layer is depicted below in Fig. 2.

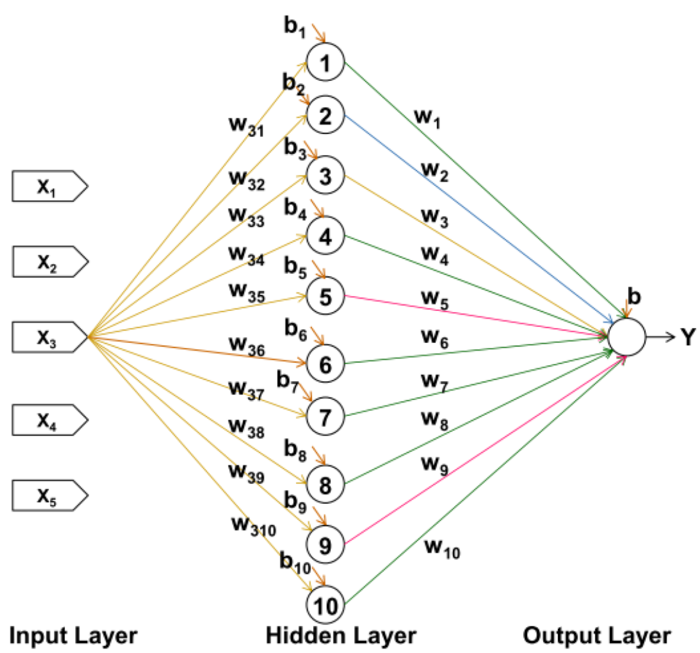

Figure 2—Typical connections from one input to hidden layers

Conversely, each hidden node receives contributions from all five inputs (Fig. 3).

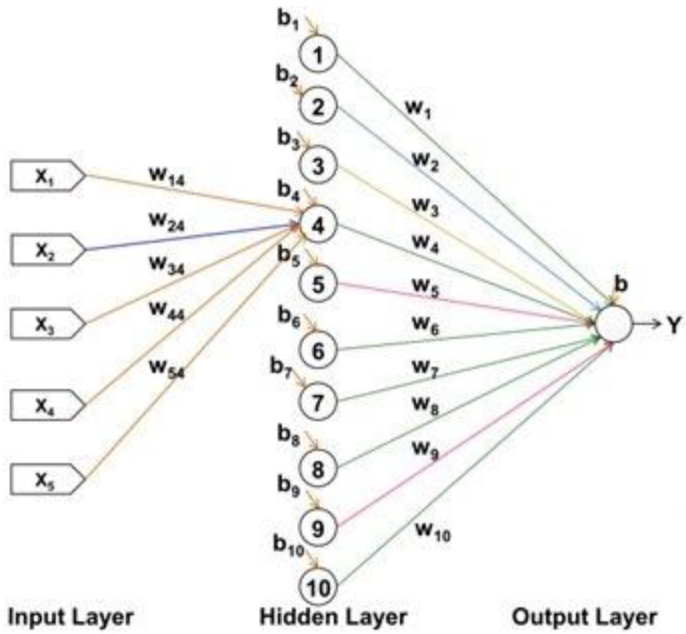

Figure 3—Typical connections from all inputs to one hidden layer node

Hidden Layer. To illustrate further, the computation, a typical connection of jth node (neuron) at the hidden layer from input to output layer is depicted in below Fig. 4:

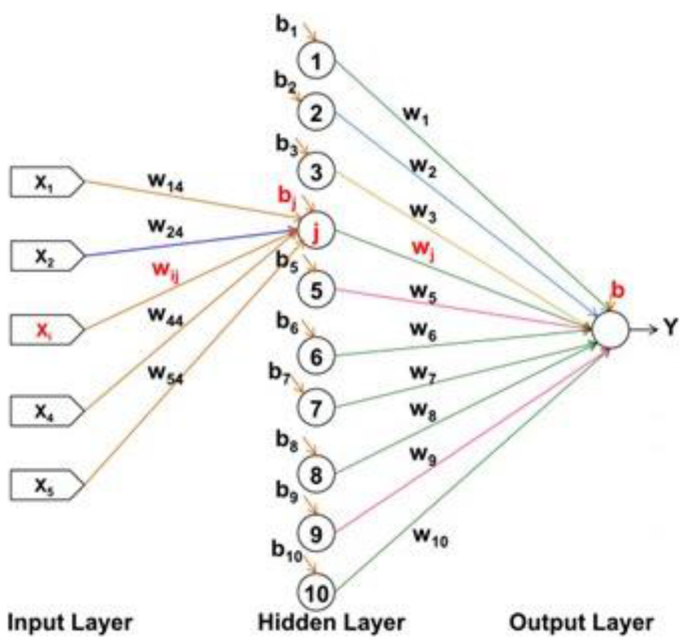

Figure 4—Typical connections from all inputs to one hidden layer node

At each hidden node (say the jth neuron), two operations occur as depicted in Fig. 5:

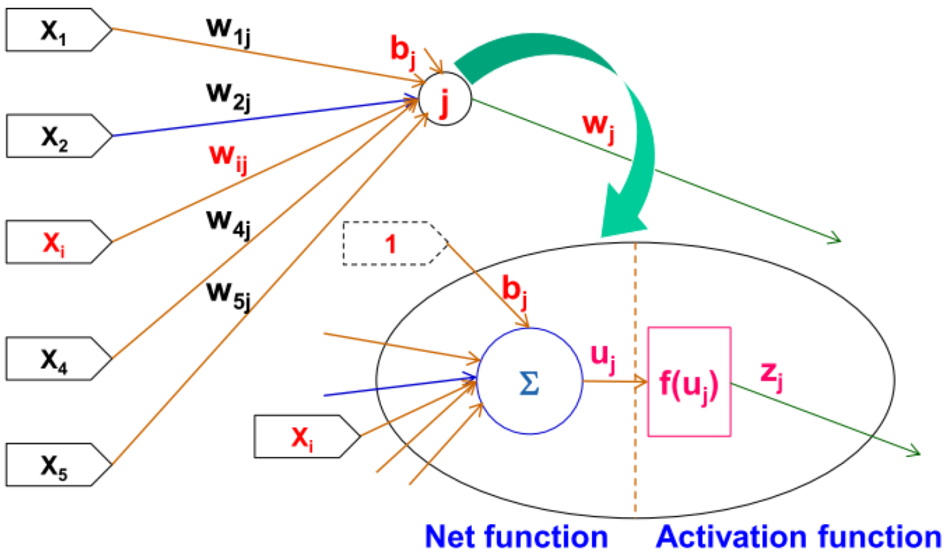

Figure 5—Typical computations at j^{th} node in hidden layer

1. **Net Function:** Inputs to the neuron are weighted according to their weight values (w_{ij}) & combined (typically summed) along with input associated to bias.

   Such mathematical operation for j^{th} neuron in the hidden layer it can be expressed as:

$$u_j = b_j + \Sigma\left(w_{ij}X_i\right) \quad \text{(Eqn. 1)}$$

2. **Activation Function:** This weighted sum is then further modified by what is known as an activation function f(u) (or transfer function) to create an output value z from the subject neuron.

$$z_j = f\left(u_j\right) \quad \text{(Eqn. 2)}$$

   In the literature, pertaining to applications of Artificial Neural Networks in Chemical Engineering, it is noted that simple bounded differential function such as the sigmoid function, having non-linearity, is usually considered as Activation Function (Himmelblau, D.M. 2000). Literature shows that sigmoid functions are widely applied in process ANN models (Cybenko, 1989; Himmelblau, 2000; Al-Otaibi et al., 2005; Arce-Medina and Paz-Paredes, 2009). Accordingly, a logistic sigmoid was adopted here. Thereby, activation function eqn. (3.2) becomes:

$$z_j = f\left(u_j\right) = 1/\left(1 + e^{-uj}\right) \quad \text{(Eqn. 3)}$$

   This output from jth neuron at the hidden layer, which shall serve as an input to neuron at the output layer.

Output Layer. The output of neurons in single hidden layer serves as the input to the neuron in the output layer. Neuron in output layer shall also apply two functions, as discussed above for neuron in hidden layer. However, it is customary to apply different activation functions in the output layer from those for the hidden layers. Also, in literature, linear transfer function is found as activation function for neuron in output layer (Rafiq, M. et al. 2001). As a result, output of the neural network, in the present work becomes:

$$Y = b + \Sigma\left(w_j z_i\right)$$
$$\text{where, } z_j = f\left(u_j\right) = 1/\left(1 + e^{-uj}\right) \quad \text{(Eqn. 4)}$$
$$u_j = b_j + \Sigma\left(w_{ij}X_i\right)$$

Model Parameters. Thus, the following 71 parameters are to be determined for solving the model equation for the proposed ANN model explained above (with 5 inputs, 1 output, 10 neurons in the single hidden layer):

Weights:
w_{ij} ($5 \times 10 = 50$ nos.)
w_j (10 nos.)
Biases:
b (1 no.), b_j (10 nos.)

Training of the ANN. In general, after an ANN is created, it needs to "learn" the implicit relationship between the inputs (independent process variables) and the output (the dependent process variable) from known datasets, through the process termed as "training". In the present work Supervised Training was used. In supervised learning / Training the network is provided with known output for every set of inputs for large known datasets, termed as training data. Connection weights & biases are assigned some values, to start the training, and adjusted through iteration to minimize the difference between predicted and known outputs, thereby allowing the network to produce output close to known output provided to the network.

Although Levenberg–Marquardt (LM) algorithm is considered as one of the most efficient algorithms for the small & median sized patterns training (Yu and Wilamowski, 2009), here the MS Excel Solver was employed. This choice underscores the paper's motivation: promoting adoption by practitioners without requiring expensive licenses or specialized software and to attain numerous benefits. As de Levie (2012) notes, the limiting factor in real-world accuracy often arises from data noise rather than algorithmic sophistication, making such implementation practical and reliable.

The error metric used was Mean Squared Error (MSE), consistent with common practice. Solver iteratively minimized MSE by adjusting the 71 parameters.

Normalization of inputs (scaled between 0–1) and de-normalization of outputs were incorporated in the spreadsheet for robustness.

Testing the ANN. To ensure generalization, the trained ANN was tested with datasets not included in training. Successful performance on these validation datasets confirmed that the trained weights and biases captured the nonlinear process behavior adequately. This step is essential for verifying the model's usability and reliability in real applications.

To emphasize, as the development of the above ANN did not involve embedding any phenomenological information in it, and because even blank cells in MS Excel can be correlated to each other through powerful mathematical formulas, the development of ANN in MS Excel combined the benefits and versatility of ANN with the universal accessibility of Excel. This resulted in a versatile model that may be populated, trained, tested, deployed, and continually updated to serve a variety of purposes, including modeling, simulation, optimization, and process control required for Operational Excellence. It was trained and tested on multiple datasets across various problem statements to demonstrate its adaptability, rapid responsiveness to input changes, and applicability to process modeling, soft sensing, and asset performance optimization. However, for the sake of brevity, this paper is limited to presenting a single case: the prediction of RVP for stabilized crude oil, as discussed in the next section.

Case study: Predicting RVP of Stabilized Crude Oil using the Versatile Model (ANN-based)

Operational Context and Problem Definition

Reid Vapor Pressure (RVP) measures the volatility of crude oil by determining the pressure exerted by vapor in equilibrium with the liquid at 37.8 °C (100 °F) under standardized conditions. It reflects the concentration of light hydrocarbons such as methane, ethane, propane, and butanes within the crude, which directly governs its behavior during handling, processing, and transportation.

In oil and gas operations, export crude specifications mandate limits on RVP to ensure safe transport and storage, maintain product quality, and comply with environmental regulations related to Volatile Organic Compound (VOC) emissions. Elevated RVP can cause excessive vaporization, leading to pressure buildup, vapor losses, and flammability hazards in tanks, pipelines, and marine vessels. Conversely, overly aggressive stabilization that drives RVP too low strips valuable light ends, reducing the economic value of the crude. RVP is therefore both a design parameter and an operational control variable. It influences the configuration of separation trains, stabilizers, and vapor recovery systems, and reliable RVP data allows operators to fine-tune operating conditions. Effective RVP control optimizes throughput, minimizes emissions, and ensures consistent product quality.

For this reason, RVP management contributes to Operational Excellence in terms of safety, environmental compliance, product quality, and profitability. Yet, often RVP management is not an activity at production facilities. A predictive model that functions as a soft sensor for RVP estimation can aid optimization by enabling continuous monitoring, proactive control, and improved asset performance, especially in line with Industry 4.0 initiatives.

Modelling Problem Formulation

The RVP of stabilized crude depends on the dynamics of the complex upstream processing system. A simplified overview of such stabilized crude production in an oilfield from Low Pressure (LP) wells is presented in Fig. 6 below:

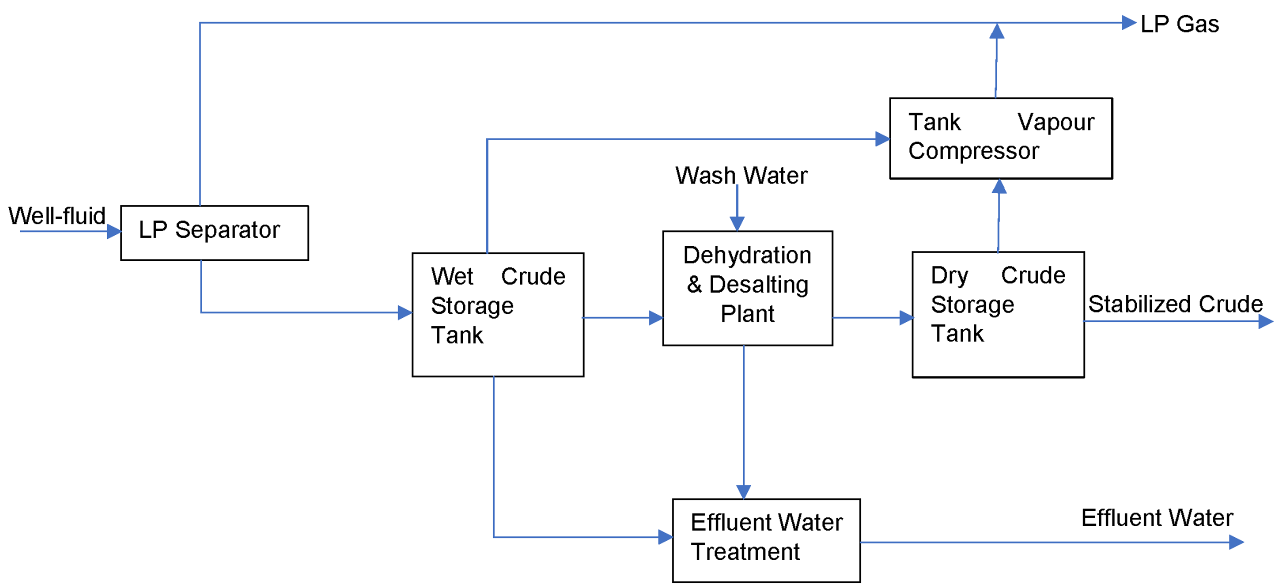

Figure 6—Simplified schematic of crude production in an oilfield

For the sake of illustration, a model for predicting RVP is built in this work using five key input variables: LP Separator Pressure and Temperature, Dry Crude Storage Tank Temperature, Well-fluid GOR and Crude API Gravity (Table 1).

Table 1—Modelling Problem Illustration

Modeling Context: Predicting RVP of Stabilized Crude Oil					
Inputs					Output
X_1	X_2	X_3	X_4	X_5	Y
LP Separator Temperature	LP Separator Pressure	Dry Crude Storage Tank Temperature	GOR	Crude API Gravity	Stabilized Crude RVP
[°C]	[psig]	[°C]	[SCF/bbl]	[API]	[psi]

Implementation in the Versatile Modeling Framework

For the case study, process data were generated using a widely adopted process simulation software capable of accurately representing thermodynamic behavior and separation processes in oil and gas facilities. It allows controlled variation of process parameters under realistic operating conditions, which is essential for developing reliable training datasets in situations where plant data may be limited or unavailable.

A total of 5256 simulated datasets were created. This dataset was then partitioned to ensure robust model training and testing. Specifically, 4382 datasets were used to populate and train the Versatile Model (ANN-based), while the remaining 874 datasets, excluded from training, were reserved for testing. This approach follows best practices in ANN development, where independent testing is required to evaluate the model's generalization capability and prevent overfitting to the training dataset. By validating unseen data, the predictive performance of the ANN can be assured under a wide range of operating scenarios. Such reliability is crucial in oil and gas applications, where decisions informed by the model directly affect safety, efficiency, and profitability.

As mentioned above, MS Excel's SOLVER was used as depicted in Fig. 7 below to train the Versatile Model (ANN-based) with the training datasets:

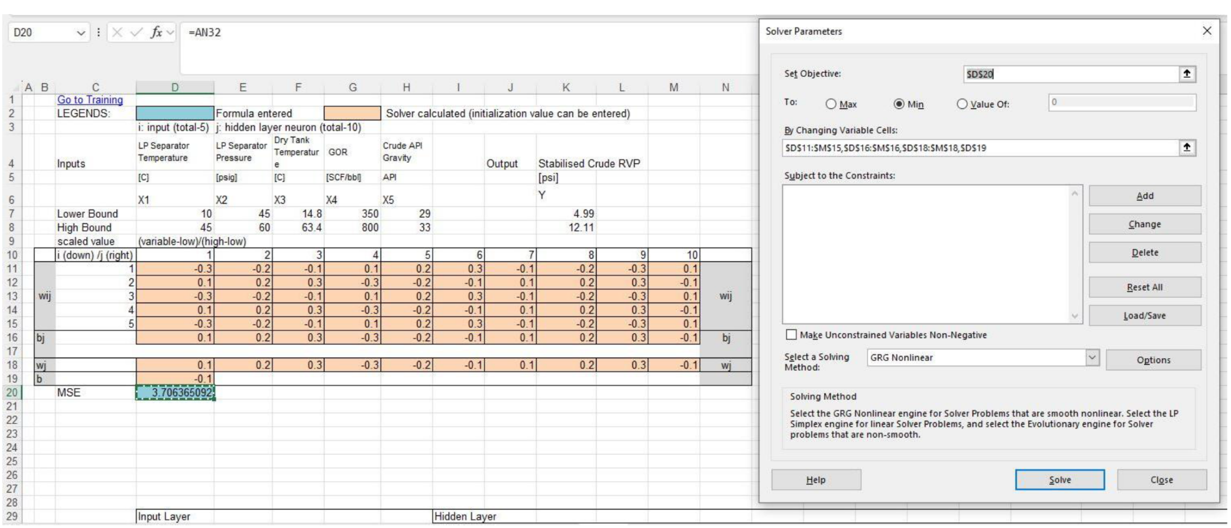

Figure 7—Versatile Model (ANN-based) Training through MS Excel's SOLVER

In this way, the integration of the state-of-the-art Process Simulation Software with the ANN-based framework demonstrates a practical pathway to develop Process Models that can also serve as soft-sensors. This is particularly relevant for achieving Operational Excellence, as it enables continuous monitoring, proactive adjustments, and asset optimization.

Results and Discussion

After training the Versatile Model (ANN-based) with the training dataset, convergence was achieved with a Mean Squared Error (MSE) of **0.00017**. For the Training Data, the comparison of the Versatile Model outputs with those from the Process Simulation Software for estimating RVP resulted in a straight-line correlation with $R^2 = 1.0$, which indicates an excellent fit. However, such a result may also reflect overfitting if the model is not validated against unseen data.

To evaluate the model's ability to generalize, the developed Versatile Model was subsequently tested against 874 independent datasets generated from the Process Simulation Software, none of which were used during the training. The predicted RVP values from the model were plotted against the corresponding Process Simulation Software outputs, as shown in Fig. 8 below:

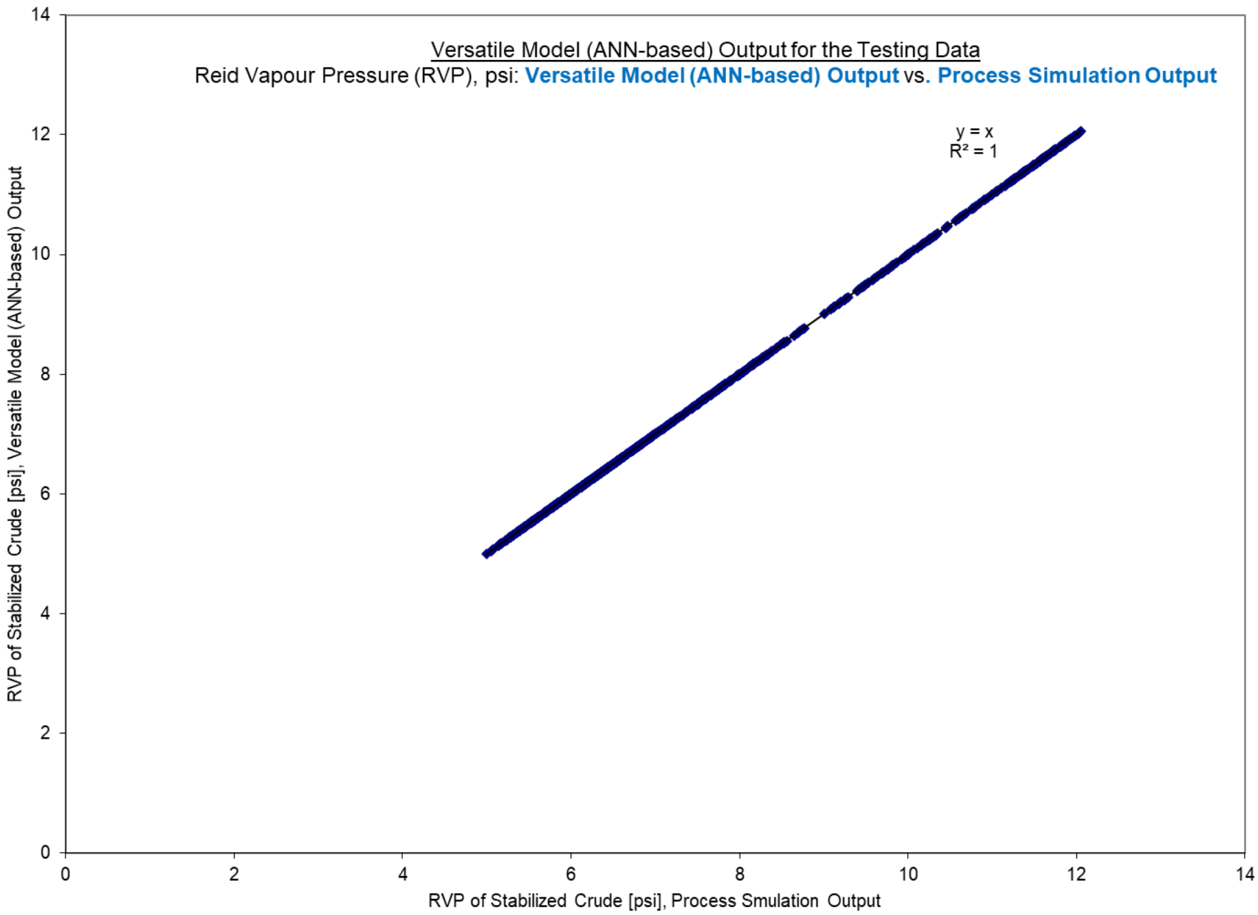

Figure 8—Evaluation of Versatile Model (ANN-based) Performance

The observed data distribution in Fig. 8, along with an $R^2 = 1$, demonstrates a perfect agreement between the Versatile Model (ANN-based) predictions and the Process Simulation Software outputs, even for data previously unseen by the Versatile Model. This confirms the model's strong generalization capability, and its performance can therefore be considered highly encouraging.

Although only one case is presented here for brevity, the Versatile Model (ANN-based) has been trained and tested on multiple datasets across different problem statements, with encouraging results.

These results demonstrate that the Versatile Model (ANN-based) can serve as a practical tool for diverse modeling needs such as optimization, forecasting, decision support, and soft sensing, contributing to facility management and asset performance optimization, and thereby supporting Operational Excellence.

To emphasize, as illustrated in the above case study, the developed Versatile Model (ANN-based), being a data-driven model implemented in spreadsheets, can be readily populated, trained, tested, and continually updated with different datasets to capture correlations among various parameters of interest across a variety of modeling contexts.

Conclusion

This paper highlights the transformative potential of AI-enabled Digital Transformation (DT) in driving Operational Excellence (OE). It recommends adopting AI-enabled solutions for asset optimization as a pathway to attaining and sustaining OE. The pivotal role of sensors and models is acknowledged, while also recognizing the challenges that act as barriers to upcoming initiatives such as Industry 4.0, particularly in contexts where the deployment of new sensors is constrained and conventional modeling tools struggle with complexity.

To address these challenges, this work presented the development of a versatile Artificial Neural Network (ANN) based process model implemented in standard spreadsheet software. Using Reid Vapor Pressure (RVP) estimation at an oilfield facility as an illustrative case, the model demonstrated how data-driven approaches can capture nonlinear relationships, provide accurate predictions, and deliver reliable insights without requiring costly platforms or extensive coding expertise. The results showed excellent agreement with process simulation outputs, confirming the model's generalization capability and underscoring its practical utility.

The Versatile Model is adaptable to a wide variety of operational contexts. It can serve roles ranging from soft sensing and process optimization to forecasting, scenario analysis, and decision support. By implementing the model in a familiar and accessible platform such as MS Excel, barriers to adoption are lowered, advanced analytics are democratized, and engineers across disciplines are encouraged to engage directly with AI-enabled solutions.

Strategically, three key insights emerge. First, AI should be viewed as an enabler rather than a replacement, complementing physical sciences and engineering practices by integrating domain expertise with data-driven insights. Second, the versatile model demonstrates broad applicability and integrative potential, acting both as a unifying framework for diverse modeling needs across the value chain and as a building block for model-enabled decision support. Third, accessible and transparent AI tools can reduce skepticism, encourage participation, and foster a culture of innovation, which is essential for embedding OE into daily practice.

In conclusion, the adoption of AI-enabled solutions should not be considered an isolated IT project but a strategic lever for asset optimization and enterprise-wide transformation. By aligning technology with people and processes, organizations can achieve sustainable improvements in safety, efficiency, and profitability, thereby advancing the vision of Operational Excellence in the era of Industry 4.0.

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